Semantic Segmentation of Remote Sensing Images by Interactive Representation Refinement and Geometric Prior-Guided Inference

Xin Li, Feng Xu, Fan Liu, *Member, IEEE*, Yao Tong, Xin Lyu, and Jun Zhou, *Senior Member, IEEE*

***Abstract*— High spatial resolution remote sensing images (HRRSIs) contain intricate details and varied spectral distributions, making their semantic segmentation a challenging task. To address this problem, it is crucial to adequately capture both local and global contexts to reduce semantic ambiguity. While self-attention modules in vision transformers capture long-range context, they tend to sacrifice local details. In this article, we propose a geometric prior-guided interactive network (GPINet), a hybrid network that refines features across encoder and decoder stages. First of all, a dual branch structure encoder with local-global interaction modules (LGIMs) is designed to fully exploit local and global contexts for feature refinement. Unlike commonly used skip connections or concatenations, the LGIMs bilaterally couple and exchange CNN features with transformer features by lossless transformation and elaborating cross-attention. Moreover, we introduce a geometric prior generation module (GPGM) that iteratively updates the randomly initialized geometric prior. Subsequently, the geometric priors are stored and used to guide feature recovery. Finally, a weighted summation is applied to the upsampled decoded features and geometric priors. By comprehensively capturing contexts and enabling lossless decoding and deterministic inference, GPINet allows the network to learn discriminative representations for accurately specifying pixel-level semantics. Experiments on three benchmark datasets demonstrate the superiority of the proposed GPINet over state-of-the-art methods. Furthermore, we validate the effectiveness of geometric priors and compare the model sizes.**

***Index Terms*— Attention bias, contextual affinity, remote sensing images (RSIs), semantic segmentation, synergistic attention.**

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I. Introduction

SEMANTIC segmentation of remote sensing images (RSIs) aims to classify and outline ground objects as interpretable semantic entities, such as buildings, roads, vegetation, and water bodies, with the purpose of extracting valuable information from Earth's surface [1], [2], [3], [4]. This task involves assigning semantic labels to image pixels or regions [5], which has facilitated various applications, including environmental monitoring [6], urban planning [7], disaster assessment [8], [9], and others.

Rather than manually selecting features based on expertise and statistical analysis, deep convolutional neural networks (DCNNs) composed of stacked convolutional layers can automatically capture diverse features, such as low-level spatial details and high-level semantics [10]. This merit has led to the evolution of DCNNs for semantic segmentation of RSIs [11]. The encoder–decoder framework, which was popularized by fully convolutional neural networks (FCNs), has emerged as the dominant baseline for semantic segmentation [12]. Since then, various FCN-variants have been investigated for RSIs. For example, SegNet gradually recovers the feature map to the original spatial size to alleviate information loss during decoding and prediction [13], [14]. Similarly, U-Net uses residual connections between the encoder and decoder to preserve features with minimal distortion [15], [16]. However, these FCN-based approaches have limitations in considering fine-grained details and context-aware information, which are essential for understanding the semantics of observed objects [17].

Inherently, vanilla CNNs focus primarily on local patterns, rather than global context. Attempting to capture comprehensive context, the atrous convolution was initially designed to enlarge the local receptive field. DeepLab V3+ integrates atrous convolutions and atrous spatial pyramid pooling for context modeling and aggregation, resulting in significant improvements in semantic segmentation [18]. Incorporating DeepLab V3+ and object-based image analysis rationale, Du et al. [19] remarkably raised the accuracy of semantic segmentation. In line with this multiscale fusion approach,

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Yue et al. [20] developed a tree-CNN block (C-Block) and a confusion matrix-based multiscale fusion module to enhance the segmentation performance of RSIs. However, this kind of multiscale interaction is unidirectional, which inevitably misses some contextual information. For small objects or regions, in particular, the grid effect and spatial details suffer more from this weakness.

In addition to convolutions, attention mechanisms (AMs) offer the alternative to model context from a global perspective [21], [22]. AMs calculate attention maps that represent pairwise similarities between arbitrary elements while ignoring positional distance. These attention maps can be used to capture long-range dependencies and complement contextual cues. Several works have focused on recalibrating channel-wise weights to pay more attention to corresponding channels of feature maps. The squeeze-and-excitation network (SENet) [23] first proposed the concept of channel attention. To capture high-order statistics, Gao et al. [24] used global second-order pooling instead of average pooling for the gathering of global information. When it comes to spatial attention, the nonlocal block was the most impressive design and laid the foundation of the self-AM [25]. Following the nonlocal routine, CCNet [26], CBAM [27], DANet [28], and OCRNet [29] were developed for semantic segmentation with competitive performance.

More recently, the transformer has shown eminent benefits in natural language processing and computer vision, demonstrating exemplary performance on many tasks. The transformer was designed based on the self-AM, which can attend to complete sequences thereby learning long-range dependencies [30], [31], [32]. Swin transformer, as a representative work, hierarchically applies attention computation with shifted windows to better extract local details and global correlations [33]. Specifically, this cascaded deployment enables the Swin transformer to inherit the advantages of ResNet [34]. However, the commonly executed tokenization process in transformers leads to the disregard of local details, reducing the discriminability of learned representations. Moreover, training a pure transformer model requires a vast amount of data to counteract the deficiency of inductive biases [35].

RSIs offer a perspective on our planet's surface with their high spatial resolution, enabling the capture of intricate details and variations in land cover. However, these images also come with inherent complexities. The spectral properties of RSIs, influenced by atmospheric conditions, create distinct spectral signatures for different land cover types, further complicating the segmentation task. In addition, the high spatial resolution introduces challenges related to the recognition of objects at varying scales and the need to differentiate fine-grained details. Moreover, RSIs are susceptible to atmospheric effects, which can distort the observed scene. These physical characteristics collectively contribute to the formidable challenge of semantic segmentation in RSIs, demanding innovative approaches that can harness the rich information while addressing these complexities. In summary, CNN-based approaches excel at preserving local perceptual information, while transformers are adept at modeling long-range contexts of visual features. To the best of our knowledge, the features extracted by CNN and transformer models are complementary. However, the interactive use of these features remains to be elaborated. In addition, from a human behavioral and cognitive perspective, the holistic perception of geometric properties can guide the understanding of semantics. Therefore, generating and incorporating geometric priors to refine learned representations can be beneficial in eliminating uncertainties around edges.

In this article, we propose a geometric prior-guided interactive network (GPINet) for the semantic segmentation of RSIs. To tackle the problems mentioned above, in the encoder stage, GPINet fuses CNN-extracted and transformer-extracted features interactively and adaptively. At the end of the encoder, a geometric prior generation module (GPGM) generates and updates geometric priors by the designed tandem cross-attention. Furthermore, a prior-guided decoder (PGD) is designed to utilize geometric information to facilitate decoding and inference, leading to accurate edge and entity delineation. The main contributions of this article are summarized as follows.

1) We introduce a local-global interaction module (LGIM) that enhances feature communication across different data types in RSIs, enabling a more nuanced understanding of spatial and spectral properties. LGIM endows the network with the ability to interpret complex land cover types by facilitating a dynamic exchange of local and global contextual information, which come from CNN-extracted and transformer-extracted features, respectively.
2) We present a GPGM that refines geometric priors by integrating spatial and geometric complexities, which helps preserve detailed features in RSIs. In our pursuit to preserve fine-grained details, we devise a feature interaction module (FIM) that sequentially organizes cross-attention and self-attention. Therefore, GPGM simultaneously refines decoded features and updates geometric priors simultaneously.
3) We propose a novel PGD to facilitate the feature recovery and dense prediction of semantic segmentation of RSIs. The proposed PGD strategically merges upsampled features with GPGM-generated priors. By highlighting the importance of geometric information in the inference process, our approach yields more precise and reliable segmentation outcomes.
4) We assemble these components into the GPINet, designed with a keen awareness of the spectral, spatial, and geometric characteristics of RSIs. GPINet is meticulously evaluated on three representative benchmarks [36], [37], [38], including aerial and satellite images. The experimental results underscore the effectiveness of GPINet compared with several state-of-the-art approaches. Furthermore, we validate the effects of PGD.

The rest of this article is organized as follows. Section II retrospects the related works of semantic segmentation of RSIs and transformer in RSIs processing. Section III describes the proposed GPINet in detail. Section IV presents the experiments and results with discussions on the pros and cons

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of models. Finally, Section V summarizes the findings of this study and draws the conclusion.

II. Related Works

A. Semantic Segmentation of RSIs

Semantic segmentation of RSIs involves labeling every pixel in images acquired from satellite or aerial platforms with land cover or land use types [39]. Due to the wide geospatial coverage and limited spatial details of RSIs, most neural networks developed for specific ground targets cannot be directly applied to RSIs. Furthermore, the variation in spectral, spatial, and temporal domains adds ambiguity to the assignment of semantic labels. Therefore, generic models that consider the properties and diverse scenarios of RSIs are necessary [40], [41].

ResUNet-a is an extension of U-Net, which integrates residual connections and attention modules for feature extraction and context modeling in RSI segmentation [42]. ResUNet-a is a combination of several advanced techniques in computer vision, highlighting the potential of merging different techniques for RSI interpretation. LANet [43] invents a patch-wise attention module to preserve the local details before multilevel fusion. Similarly, Li et al. [44] proposed shallow layer spatial attention and deep layer channel attention modules for the semantic segmentation of satellite images. In line with the hybrid design, SCAttNet [45], HMANet [46], and HCANet [47] capture and fuse channel and spatial attentional contexts before inference. Moreover, Ding et al. [48] proposed a WiCoNet that adopts two CNNs to extract features from local and global image levels, respectively. Accordingly, it is helpful to capture long-range dependencies and inject such contexts to boost the semantic segmentation of RSIs [49], [50].

Recently, the transformer, which sequentially captures contextual dependencies between arbitrary elements, has shown great potential for computer vision tasks [35]. SETR [51] formulates semantic segmentation as a sequence-to-sequence prediction task and encodes an image as a series of patches, resulting in competitive performance on natural images. Similarly, the segmenter model [52] builds a segmentation transformer based on a vision transformer. Before fine-tuning the semantic segmentation task, the segmenter model is pretrained on an image classification dataset. The class masks are then generated by a simple point-wise linear decoder.

Despite the substantial improvement, the tokenization in the transformer deteriorates object structures and ignores inner-patch local details. In contrast, CNNs hierarchically ensemble local features. Motivated by the feature refinement paradigm, we retrospect to facilitate communication between CNN and transformer features. This interaction procedure is deemed to cooperate CNN with the transformer beyond simple fusion. Furthermore, to better exploit and leverage structural information, we propose to generate geometric priors and inject them into the decoder and classifier as guidance. By incorporating geometric priors, the inference is constrained by more available cues without sacrificing convergence efficiency or requiring extra supervision.

B. Transformer in Remote Sensing

The transformer has shown great potential in modeling distance-free contextual information and has been successfully applied to RSI processing [53]. For hyperspectral image classification, Yang et al. [54] proposed a multilevel spectral-spatial transformer that achieved impressive results on two real datasets. Moreover, He et al. [55] employed bidirectional encoder representations from transformers for hyperspectral image classification, setting a new state-of-the-art performance. In the field of pansharpening, the transformer has also demonstrated excellent performance. Deng et al. [56] developed a pyramid shuffle-and-reshuffle transformer that leverages the strong correlation among different patches in RSIs and the complementary information among patches with high similarity.

In the realm of high-resolution RSIs, transformers have been utilized in scene classification [57], object detection [58], and change detection [59]. In the context of semantic segmentation of RSIs, Zhang et al. [60] proposed a hybrid network, where the Swin transformer is combined with a U-shape decoder. Likewise, He et al. [61] parallelly deploys the Swin transformer and CNN backbone for feature encoding. Spatial interaction is implemented to enhance occluded objects' representations. Moreover, with the relational aggregation module, the constructed ST-UNet brings significant improvements to the ISPRS benchmarks.

Unlike the aforementioned approaches, the proposed GPINet in this article integrates the features extracted by CNN and transformer through cross-attentive coupling and aggregation, aligning and connecting the misaligned features during the interaction. This design allows the network to adjust and bridge the misalignment between the two components. In addition, GPINet utilizes learnable geometric priors to compensate for the lack of structural information, which is updated during training. Cooperated with PGD, GPINet further highlights the influence of geometric priors, which greatly contributes to the accuracy of predictions.

III. Method

To effectively tackle the challenges posed by RSIs, our proposed GPINet integrates a series of carefully designed modules, each tailored to address specific characteristics of RSIs. The LGIMs within GPINet are strategically engineered to leverage the spatial-spectral properties of RSIs. By facilitating bilateral coupling and the exchange of CNN features with transformer features through lossless transformations and cross-AMs, LGIMs enable the network to capture spatial-spectral variations more effectively, ensuring accurate delineation of land cover types with distinct spatial-spectral signatures. To be specific, the core idea behind this transformation is to convert the spatially rich feature maps extracted by the CNN into a sequence of flattened 1-D tokens without any information loss. This is crucial because the transformer architecture, which is adept at capturing global dependencies, processes input sequentially and lacks innate spatial awareness. Our method preserves spatial information by embedding it into these tokens via positional encodings.

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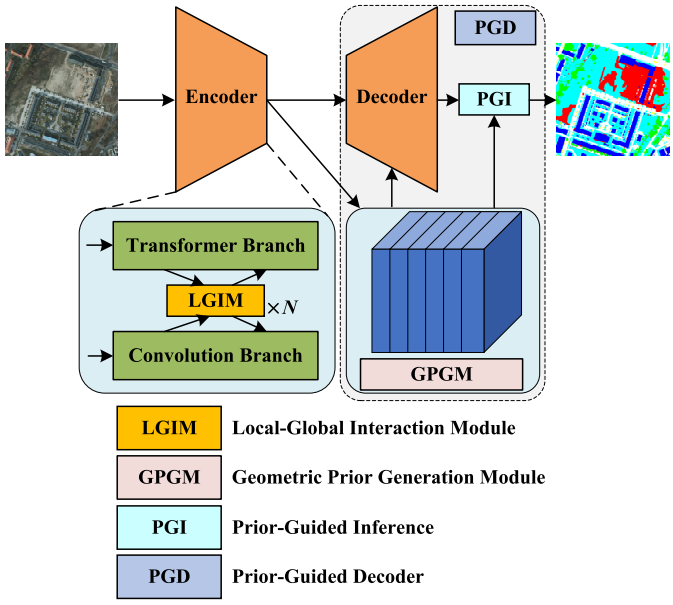


Fig. 1. Overview of GPINet.

These encodings ensure that the transformer can consider the original spatial relationships of the features within the image. Once transformed, these features serve as the query in a cross-AM within the LGIM. The cross-attention facilitates an exchange of information between the local detail captured by the CNN features and the global context captured by the transformer features. This is realized by computing attention weights that reflect the significance of each feature in the context of the whole image, allowing the model to make informed predictions based on both local and global information.

Furthermore, our GPGM accounts for the geometric considerations inherent in RSIs. By iteratively updating and incorporating geometric priors, this module guides the feature recovery process, enabling precise segmentation of objects in varying orientations and terrains. These design choices are grounded in our understanding of the spectral, spatial, and geometric characteristics of RSIs, and collectively, they empower GPINet to excel in the challenging task of semantic segmentation in RSIs.

A. Overview

As presented in Fig. 1, the proposed method is built on an encoder–decoder architecture, combining an interactive feature encoder and a geometric prior guided decoder. The encoder stage deploys LGIMs to allow effective communication between the CNN and transformer branches, ensuring the flexible interaction and aggregation of local and global contexts. To enhance the accuracy of inference, the GPGM generates geometric priors that are injected into decoded features before dense prediction. This integration of comprehensive contexts and geometric priors makes GPINet robust against high intraclass variance and interclass similarity. Detailed descriptions are provided in Sections III-B–III-D.

B. Local-Global Interactive Encoder

To handle the complexity and variability of RSIs, it is crucial to capture both local and global contexts for generating informative features. To achieve this goal, we propose a dual-branch encoder, called local-global interactive encoder (LGIE), that simultaneously employs CNN and transformer networks. Specifically, the LGIM between two branches is embedded to interact and refine their learned representations with comprehensive contexts. This module is essential for producing discriminative representations that can effectively capture the complex scenarios and variable scales and appearances of RSIs.

Fig. 2 presents the details of LGIM. Initially, the input image is fed into a convolution layer and a patch partition operator, respectively, generating the CNN feature and the transformer feature. Prior to LGIM, the input image with a size of $256 \times 256 \times 3$ is initialized by a 7×7 convolution with stride 2. Meanwhile, the tokenization produces the transformer feature where a patch embedding is with $4 \times 4 \times 3$. Notably, the initial CNN feature is with a size of $128 \times 128 \times 64$, and the transformer feature is with 4096×48, where 4096 and 48 are the numbers of patches and dimensionality of a single patch, respectively. It can be observed that the input features are with different sizes and dimensionalities of LGIM. Therefore, it is necessary to align one with another to mitigate inconsistency in interaction.

Let $F_{\mathrm{CNN}} \in \mathbb{R}^{C_c \times H \times W}$ and $F_{\mathrm{TRA}} \in \mathbb{R}^{N \times C_t}$ represent the input CNN feature and the transformer feature of LGIM, respectively. F_{CNN} is first transformed by the C2T module as shown in Fig. 2(b). Formally

$$F_{\mathrm{C2T}} = \mathrm{LN}(\mathrm{Reshape}(\mathrm{AP}(\mathrm{Conv}_{1\times1}(F_{\mathrm{CNN}})))) \tag{1}$$

where $F_{\mathrm{C2T}} \in \mathbb{R}^{(\mathrm{HW}/4)\times C_t}$ is the transformed CNN feature of F_{CNN}, AP denotes 2×2 average pooling, C_t represents the dimenionality of F_{C2T}, and LN represents layer normalization. Moreover, the number of feature vectors of F_{C2T} is equal to F_{TRA} ($(\mathrm{HW}/4) = N$). As previously discussed, C2T projects CNN feature to transformer style in a simple yet effective manner. Consequently, with the same size, F_{C2T} can be treated as the query branch to respond to the key branch of transformer feature F_{TRA}.

After alignment, the local-global interaction is implemented in a cross-attention manner by convolution to the transformer interaction module (C2TI). As shown in Fig. 2(d), C2TI employs F_{C2T} as query to response F_{TRA}, injecting local details into global context

$$F_{\mathrm{TRA}}^{\mathrm{C2TI}} = \mathrm{Softmax}\left(\frac{(F_{\mathrm{C2T}}W_q)(F_{\mathrm{TRA}}W_k)^T}{\sqrt{C_t}}\right)(F_{\mathrm{TRA}}W_v) \tag{2}$$

where W_q, W_k, and $W_v \in \mathbb{R}^{C_t \times C_t}$ are learned projection matrices and $F_{\mathrm{TRA}}^{\mathrm{C2TI}} \in \mathbb{R}^{N \times C_t}$ is the refined F_{TRA}. Afterward, $F_{\mathrm{TRA}}^{\mathrm{C2TI}}$ is forwarded to the next transformer block (T-Block).

The T-Block, in this study, is composed of a patch merging module and two successive Swin T-Blocks [33]. As formulated below

$$F_{\mathrm{TRA1}} = \mathrm{ST}\left(\mathrm{PM}\left(F_{\mathrm{TRA}}^{\mathrm{C2TI}}\right)\right) \tag{3}$$

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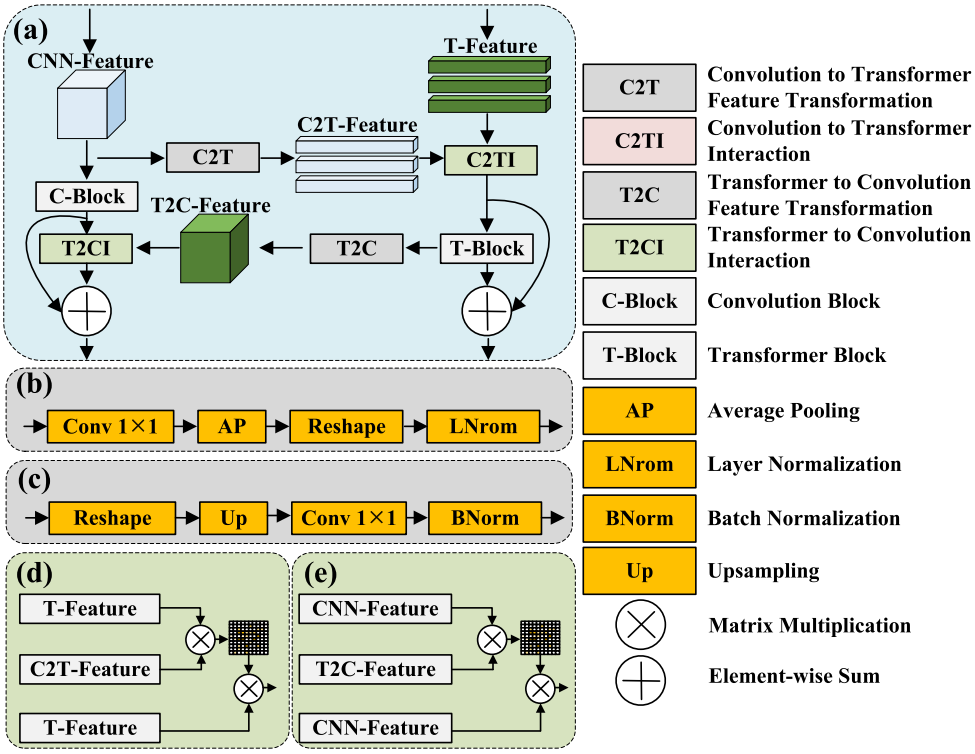


Fig. 2. Details of LGIM. (a) Pipeline of LGIM. (b) Pipeline of C2T. (c) Pipeline of T2C. (d) Pipeline of convolution to transformer interaction. (e) Pipeline of T2CI.

where PM() and ST() represent the patch merging and Swin T-Block, respectively. The output F_{TRA1} is then fed into the next stage and used to optimize the CNN feature. Similar to the forward process, F_{TRA1} is transformed to CNN-like feature by the transformer to convolution module (T2C) [see Fig. 2(c)], followed by transformer to convolution interaction (T2CI) [see Fig. 2(e)]. The T2C module can be represented as

$$F_{\text{T2C}} = \text{BN}(\text{Conv}_{1\times1}(\text{Up}(\text{Reshape}(F_{\text{TRA1}})))) \tag{4}$$

where Up() and BN() are upsampling and batch normalization, respectively. In response to the CNN feature, T2CI is denoted as

$$F_{\text{TRA1}}^{\text{T2CI}} = \text{Softmax}\left(\frac{(F_{\text{T2C}}W_q)(F_{\text{CNN1}}W_k)^T}{\sqrt{C_{c1}}}\right)(F_{\text{CNN1}}W_v) \tag{5}$$

where $F_{\text{CNN1}} \in \mathbb{R}^{(H/2)\times(W/2)\times C_{c1}}$ is the output of C-Block and F_{T2C} is aligned with F_{CNN1}. Finally, the output features can be obtained by two element-wise summation operators

$$F_{\text{CNN1}}^{R} = F_{\text{TRA1}}^{\text{T2CI}} \oplus F_{\text{CNN1}} \tag{6}$$

$$F_{\text{TRA1}}^{R} = F_{\text{TRA}}^{\text{C2TI}} \oplus F_{\text{TRA1}} \tag{7}$$

where F_{CNN1}^{R} and F_{TRA1}^{R} are refined features by LGIM.

LGIM is designed to communicate CNN-extracted features with transformer-extracted features and exchange informative details, enabling the encoder to learn more comprehensive contexts. Unlike other methods, our proposed interaction is simple yet effective in collecting sufficient contextual cues.

C. Geometric Prior Generation Module

As previously analyzed, ground objects are visually ambiguous around edges or between similar yet inhomogeneous land cover types. This salient property is intrinsically caused by fragmentary geometric information. To address this issue, we propose a solution by generating and updating geometric priors, which can be used as guidance for decoding features and predicting pixel-level semantics. This approach allows us to better store and exploit geometric information.

As shown in Fig. 3, the randomly initialized geometric prior is with a size of $F_{\text{gp}} \in \mathbb{R}^{H_g\times W_g\times C_{\text{cls}}}$, while the output feature of encoder is denoted as $F_{\text{enc}} \in \mathbb{R}^{H\times W\times C}$, where H, W, and C are the height, width, and channel of F_{enc}, respectively, and H_g, W_g, and C_{cls} are the height, width, and channel of F_{gp}, respectively. Notably, C_{cls} is equal to the number of classes. When $(H, W) \neq (H_g, W_g)$, the geometric priors could be resized by interpolation to (H and W).

At the beginning of the FIM, F_{gp} is multiplied by F_{enc} to produce a cross-attention map (softmax omitted in Fig. 3) as follows:

$$\text{AttnMap} = \text{Softmax}\left(\frac{(F_{\text{enc}}W_q)(F_{\text{gp}}W_k)^T}{\sqrt{C}}\right) \tag{8}$$

where AttnMap $\in \mathbb{R}^{C_{\text{cls}}\times C}$ involves the correlations between encoded features and geometric priors. Then, F_{enc} and F_{gp} are refined by AttnMap, respectively. They are $F_{\text{enc}}^{\text{ca}} \in \mathbb{R}^{H\times W\times C_{\text{cls}}}$ and $F_{\text{gp}}^{\text{ca}} \in \mathbb{R}^{H\times W\times C}$, where $F_{\text{enc}}^{\text{ca}}$ and $F_{\text{gp}}^{\text{ca}}$ are refined F_{enc} and F_{gp}. After concatenation, $F_{\text{concat}}^{\text{ca}} \in \mathbb{R}^{H\times W\times(C_{\text{cls}}+C)}$ is input to a self-attention module. Formally

$$F_{\text{concat}}^{\text{sa}} = \text{Softmax}\left(\frac{(F_{\text{concat}}^{\text{ca}}W_q)(F_{\text{concat}}^{\text{ca}}W_k)^T}{\sqrt{C}}\right)(F_{\text{concat}}^{\text{ca}}W_v) \tag{9}$$

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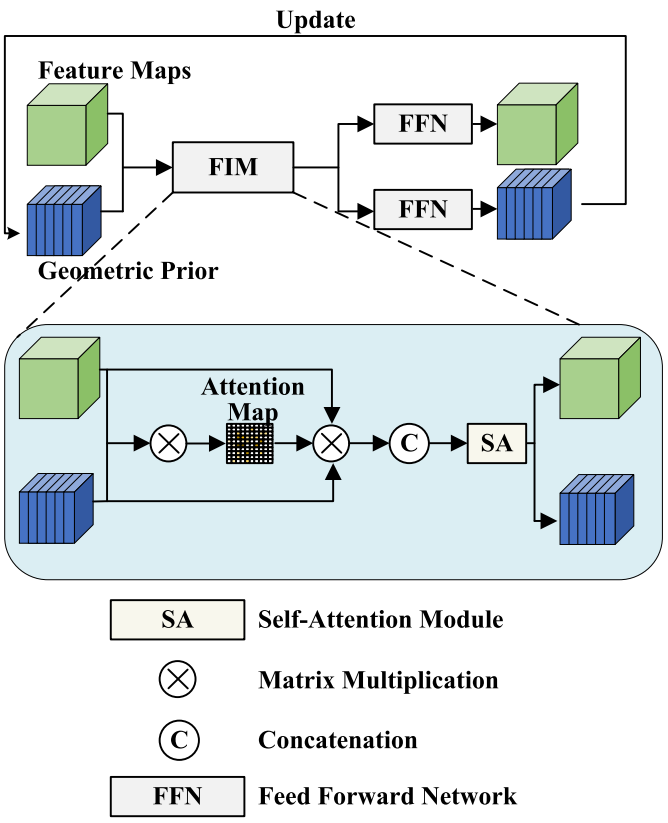


Fig. 3. Details of GPGM.

where the generated self-attention map is with a spatial size of $(C_{\text{cls}}+C)\times(C_{\text{cls}}+C)$. Finally, $F^{\text{sa}}_{\text{concat}}$ is split to form F_{encr} and F_{gp} along channel dimension.

As introduced above, the purpose of GPGM is to iteratively update geometric priors by incorporating more useful contextual information from the feature encoder. This tandem design adopts cross-attention and self-attention as the core units to guarantee an appropriate interaction and refinement.

D. Prior-Guided Decoder

The geometric priors are generated and updated to gather and store the dataset-level geometric information. To further utilize it, as shown in Fig. 4, PGD harnesses geometric priors as guidance for feature decoding and inference.

First of all, we shrink the channel of F_{encr} to C_{cls} by a 1×1 convolution layer. Moreover, matrix multiplication is applied to inject geometric priors. Then, an element-wise summation refines decoded features. Therefore, the feature decoding can be formed as

$$F^{\text{pgd}}_{\text{encr}} = F_{\text{gp}} \times \text{Conv}_{1\times 1}(F_{\text{encr}}) \oplus F_{\text{gp}} \tag{10}$$

where $F^{\text{pgd}}_{\text{encr}}$ is the decoded features with prior guidance. Moreover, a weighted summation function is applied to aggregate upsampled features and geometric priors as follows:

$$F_{\text{predict}} = \alpha \text{Up}\left(F_{\text{gp}}\right) + \beta s \text{Up}\left(F^{\text{pgd}}_{\text{encr}}\right) \tag{11}$$

where α and β are coefficients (practically both set to 0.5 in this study).

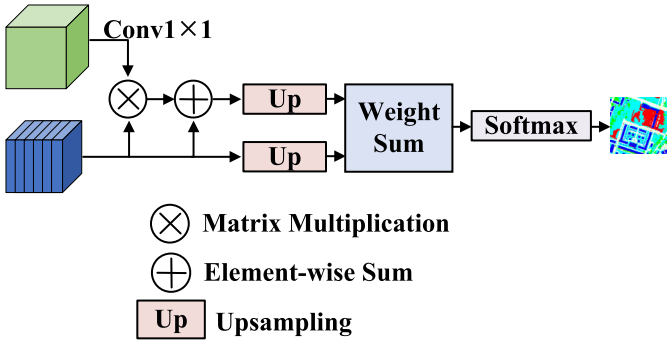


Fig. 4. Details of PGD.

Finally, the proposed PGD densely predicts pixel-level semantics using the softmax function. By utilizing the implicit correlations between decoding features and the stored geometric priors, the PGD can learn discriminative representations without the need for complex computations. In general, our network adopts an encoder–decoder framework complemented by sophisticated representation refinement modules. Cross-entropy loss is commonly employed in semantic segmentation tasks to effectively manage class imbalances by concentrating on the probability attributed to the correct class [62]. Hence, we selected cross-entropy loss for its proven effectiveness in multiclass tasks, especially in cases where the output is construed as a probability distribution.

E. Convergence Analysis

In this section, we introduce a theoretical framework to analyze the convergence properties of the proposed neural network. We specifically focus on the cross-entropy loss function, which is defined for a multiclass classification problem as follows:

$$L(\theta) = -\sum_{i=1}^{m}\sum_{c=1}^{C} y_{\text{ic}} \log(p_{\text{ic}}(\theta)) \tag{12}$$

where L is the loss function, θ represents the parameters of the network, m is the number of samples, C is the number of classes, y_{ic} is the binary indicator of the ground truth class for the ith sample and class, and $p_{\text{ic}}(\theta)$ is the predicted probability given the parameters θ.

To establish the convergence rate, we examine the Lipschitz continuity of the gradients of the cross-entropy loss. The gradient with respect to the parameters θ is given by

$$\nabla_\theta L(\theta) = -\sum_{i=1}^{m}\sum_{c=1}^{C}(y_{\text{ic}} - p_{\text{ic}}(\theta))x_{\text{ic}}. \tag{13}$$

Assuming that the feature vectors x_{ic} are bounded, i.e., $\|x_{\text{ic}}\|_2 \le X_{\max}$ for some constant $X_{\max}$, we can assert that the gradients are Lipschitz continuous with a Lipschitz constant L, such that

$$\left\|\nabla_\theta L(\theta) - \nabla_\theta L\left(\theta'\right)\right\|_2 \le L\left\|\theta - \theta'\right\|_2 \tag{14}$$

for all θ, θ' in the parameter space, where the Lipschitz constant L is proportional to $X_{\max}$ and the maximum possible value of $(y_{\text{ic}} - p_{\text{ic}}(\theta))$.

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TABLE I
DATASETS

Datasets	Vaihingen		LoveDA
Bands used	R, G, B	R, G, NIR	R, G, B
Spatial resolution	9 cm	5 cm	0.3 m
Number of available images	33	38	4191
Spatial size	About 2494×2064	6000×6000	1024×1024
Imaging sensors	Airborne	Airborne	Satellite
Sub-patch size	256×256	256×256	256×256
Training set (number of iamges)	14	17	1677
Validation set (number of iamges)	2	2	419
Test set (number of iamges)	17	19	2095

Under this assumption, and given a learning rate α satisfying $0 < \alpha \leq (1/L)$, we apply the results of stochastic gradient descent (SGD) convergence, which ensures that the expected value of the loss function converges to a neighborhood of the global minimum at a rate of $\mathcal{O}((1/(T)^{1/2}))$, where T is the number of iterations

$$\mathbb{E}[L(\theta_T)] - L\left(\theta^*\right) \leq \frac{G}{\sqrt{T}} \tag{15}$$

with G being a constant dependent on the Lipschitz constant and the variance of the gradient estimates.

This theoretical analysis supports the empirical observations of satisfactory convergence rates in our experiments (see in Section IV-H). It confirms that under reasonable assumptions, the network parameters converge toward a set of optimal parameters that minimize the cross-entropy loss, thereby validating our choice of the loss function and learning rate for the problem at hand.

IV. Experiments and Discussions

A. Datasets

In this study, we have randomly allocated the images in the training and validation sets in nearly a 1:1 ratio. Details can be found in Table I. Our dataset compilation consists of three distinct sets: ISPRS Vaihingen, ISPRS Potsdam, and LoveDA, comprising aerial and satellite images characterized by varying resolutions. The subsequent subsections offer a comprehensive introduction to each of these datasets.

1) ISPRS Vaihingen Dataset: The ISPRS Vaihingen dataset was acquired from an aerial platform with a very high resolution of 9 cm. The ground truth is labeled with pixel-level land cover types, in which clutter is defined as the background class. There are 33 images with an average spatial size of 2494×2064 pixels available. Regarding the input images, we adopt red (R), green (G), and blue (B) bands. We divide the dataset into a training set, a validation set, and a test set, with the size of 14, 2, and 17 images, respectively.

2) ISPRS Potsdam Dataset: The ISPRS Potsdam dataset has a spatial resolution of 5 cm. The same semantic labels as the ground truth of the Vaihingen dataset are defined. Each image has a spatial size of 6000×6000 pixels, with the R, G, and near-infrared (NIR) bands used. We divide the dataset into a training set, a validation set, and a test set, with the size of 17, 2, and 19 images, respectively.

TABLE II
EXPERIMENTAL SETTINGS

Items	Settings
Learning strategy	SGD
Initial learning rate	0.02
Loss function	Cross-entropy
Max epoch	500
GPU memory	48 GB
Sub-patch size	256×256
Batch size	32

3) LoveDA Dataset: The LoveDA dataset intrigued a new challenge in semantic segmentation of large-scale satellite images with a spatial resolution of 0.3 m. Acquired from the Google Earth Platform, LoveDA covers a total area of over 536 km^2. It includes both rural and urban areas of three cities, i.e., Nanjing, Changzhou, and Wuhan. Each image in the dataset has a spatial size of 1024×1024 pixels. In this study, we adopted 2522 images for training, 834 images for validation, and 835 images for testing. The class distribution is imbalanced, and congeneric objects vary in scale, size, and surface type, making LoveDA an even more challenging dataset.

B. Experimental Settings

The proposed GPINet and the compared semantic segmentation methods were implemented using PyTorch on a Linux OS with an NVIDIA A40 GPU. Data augmentations, such as random flipping and cropping operations, were applied to all datasets and networks, as shown in Table II. The initial learning rate and maximum epoch were fixed at 0.02 and 500, respectively. Moreover, we adopt SGD as the optimizer. The learning strategy and momentum are set as poly decay and 0.9, respectively. The model parameter file with the lowest validation loss was saved.

We selected ten methods for comparison. These included FCN-8s [12], U-Net [15], DeepLab V3+ [18], and DANet [27], which are representative baselines for semantic segmentation task. DANet, in particular, introduces a dual AM for semantic segmentation. In addition, we included recently proposed methods for semantic segmentation of RSIs, i.e., RAANet [63], MACUNet [64], HCANet [47], SCAttNet [45], A2FPN [65], and LANet [43].

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TABLE III
RESULTS ON THE ISPRS VAIHINGEN DATASET. CLASS-WISE $F1$-SCORE, AF, OA, AND MIOU ARE LISTED, WHERE THE BOLD TEXT INDICATES THE BEST RESULTS

Methods	Impervious surfaces	Building	Low vegetation	Tree	Car	AF	OA	mIoU
FCN-8s [12]	84.08	73.61	65.09	79.97	38.76	68.30	66.67	62.38
U-Net [15]	84.67	86.13	69.49	82.71	41.89	72.98	71.24	66.66
DeepLabV3+ [18]	87.99	87.80	72.72	85.55	47.37	76.29	74.47	69.69
DANet [27]	91.57	90.37	80.01	88.15	58.50	81.72	79.78	74.65
RAANet [63]	90.09	89.72	76.22	86.13	48.55	78.14	75.33	70.49
MACU-Net [64]	89.52	90.11	74.32	87.15	62.71	80.76	77.86	72.86
HCANet [47]	**94.29**	**96.20**	83.33	92.38	88.86	91.01	88.84	81.13
SCAttNet [45]	90.44	91.63	81.22	81.49	71.54	83.26	81.52	75.29
A2FPN [65]	92.99	95.17	80.98	89.25	87.62	89.20	87.07	81.06
LANet [43]	93.81	95.70	82.93	91.94	88.44	90.56	87.48	81.72
GPINet (ours)	93.80	95.98	**86.69**	**92.39**	**89.17**	**91.58**	**89.40**	**82.57**

C. Evaluation Metrics

In this study, we use commonly adopted evaluation metrics to measure the performance of the predicted results on the test set, including class-wise F_1 score (F_1), average F_1 score over all classes (AF), overall accuracy (OA), and mean intersection over union (mIoU). $F1$ score is the harmonic mean of precision and recall, which evaluates the tradeoff between false positives and false negatives. OA measures the number of correctly classified pixels against all pixels. mIoU is calculated as a global metric to evaluate accuracy. The equations of $F1$, OA, and IoU are presented, respectively,

$$F_1 = 2 \cdot \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}} \tag{16}$$

$$\text{OA} = \frac{(\text{TP} + \text{TN})}{(\text{TP} + \text{TN} + \text{FP} + \text{FN})} \tag{17}$$

$$\text{IoU} = \frac{\text{TP}}{(\text{TP} + \text{FP} + \text{FN})} \tag{18}$$

where precision and recall are calculated as

$$\text{precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{19}$$

$$\text{recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{20}$$

where TP, TN, FP, and FN are the numbers of true positive, true negative, false positive, and false negative samples, respectively. mIoU is the average of class-wise IoU.

D. Comparison With the State-of-the-Art Methods

1) Results on the ISPRS Vaihingen Dataset: As shown in Table III, thequantitative comparison with state-of-the-art methods on the ISPRS Vaihingen dataset is presented. The proposed GPINet gains the best numerical performance among all compared methods. It is worth noting that our GPINet yields an 86.69 $F1$-score on the "low vegetation" class, exceeding the previous best method HCANet by more than 3. For the "tree" and "car" classes, GPINet reaches 92.39 and 89.17 of the $F1$-score, outperforming other methods. Due to the feature refinement and geometric prior guidance, GPINet attains 91.58, 89.40% and 82.57% of AF, OA, and mIoU, respectively. Such an increase in overall metrics convincingly validates the robustness and effectiveness of GPINet. In contrast, other networks, including FCN-8s, U-Net, and DeepLab V3+, struggle to address semantic ambiguity, leading to misclassification. Visualization comparisons are shown in Fig. 5. Attention-embedded methods also fall short of distinguishing easy-confused categories, such as "tree" and "low vegetation." Our GPINet incorporates local and global contexts and takes geometric priors as guidance for feature decoding, resulting in better pixel-level representations and improved segmentation results, including easy-confused objects.

2) Results on the ISPRS Potsdam Dataset: As shown in Table IV, RSI-specific methods outperform baseline encoder–decoder networks. HCANet and LANet, which incorporate multiple attention modules to refine learned representations, achieved more than 90% of OA. This suggests that modeling and leveraging comprehensive context can contribute to distinguishing objects with similar appearance and spectral properties. Therefore, our GPINet captures both local and global contexts along with multiple interactions, which mitigates feature misalignment and semantic uncertainty. In terms of numerical metrics, GPINet achieved the highest AF, OA, and mIoU over other state-of-the-art methods, reaching a peak of 93, 91.57% and 83.33%, respectively. Visualization comparison is shown in Fig. 6, and the results are in line with the numerical metrics. Referring to ground truth, GPINet-predicted labels produced the best consistency, while comparative methods failed to fully understand the intraclass variations and interclass similarities.

3) Results on the LoveDA Dataset: The LovaDA dataset encompasses both rural and urban areas of three cities with diverse building, water, vegetation, and crop types, making it more challenging. Furthermore, the lower spatial resolution of LovaDA, as compared to aerial images, reduces the availability of spatial details. As evident from Table V, all examined methods' numerical values decrease considerably. However, our GPINet outperforms the previous best method, LANet, by approximately 2, 2%, and 3% on AF, OA, and mIoU, respectively. This is caused by two aspects, i.e., comprehensive contexts and geometric priors' guidance. For water areas, an $F1$-score of more than 91 is obtained. Similarly, GPINet segments buildings and roads with an $F1$-score of more than 80 and 82. Fig. 7 shows the predicted results of random samples. GPINet displays robust scalability while being applied to satellite images in addition to aerial images.

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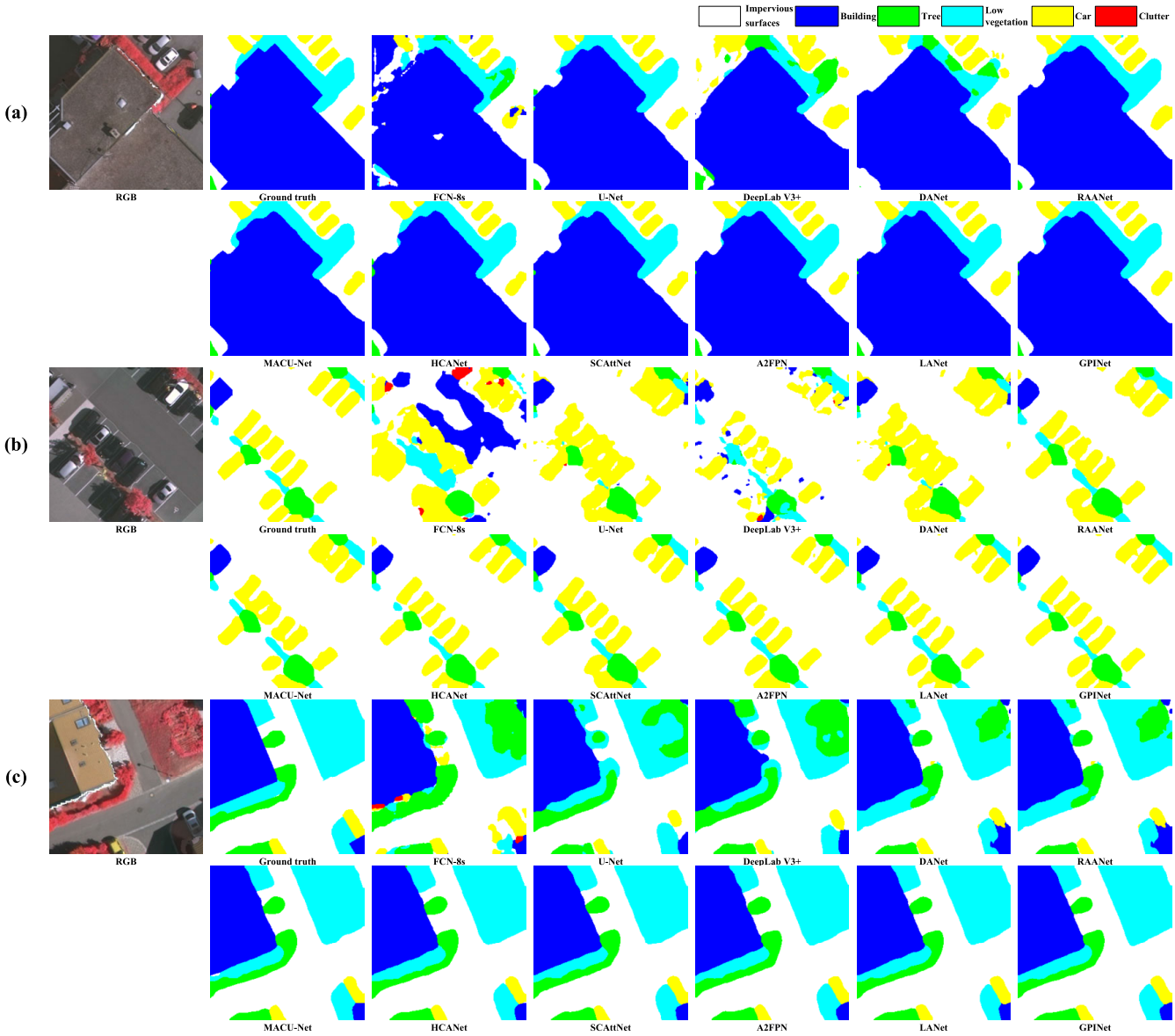


Fig. 5. Visualization comparisons on the ISPRS Vaihingen test set.

TABLE IV

RESULTS ON THE ISPRS POTSDAM DATASET. CLASS-WISE $F1$-SCORE, AF, OA, AND MIOU ARE LISTED, WHERE THE BOLD TEXT INDICATES THE BEST RESULTS

Methods	Impervious surfaces	Building	Low vegetation	Tree	Car	AF	OA	mIoU
FCN-8s [12]	85.66	74.99	66.31	81.47	39.49	69.58	68.91	64.47
U-Net [15]	87.51	89.32	73.98	87.03	48.19	77.21	75.83	70.02
DeepLabV3+ [18]	84.49	86.13	77.37	77.37	85.47	82.17	80.82	73.59
DANet [27]	86.96	92.04	80.14	79.95	89.36	85.69	83.99	77.13
RAANet [63]	90.40	95.70	87.20	81.80	77.60	86.54	85.21	77.95
MACU-Net [64]	89.16	92.76	86.80	83.02	79.25	86.20	85.90	77.99
HCANet [47]	92.88	96.90	87.25	88.15	93.88	91.81	90.67	81.92
SCAttNet [45]	91.87	96.90	85.24	87.05	92.78	90.77	89.06	80.83
A2FPN [65]	90.67	95.69	85.54	85.44	91.55	89.78	88.25	80.67
LANet [43]	92.41	96.38	86.82	87.77	93.44	91.36	90.06	81.19
GPINet (ours)	**93.78**	**97.50**	**88.15**	**88.86**	**96.70**	**93.00**	**91.57**	**83.33**

E. Effects of LGIM

Within GPINet, we have integrated the LGIMs to boost feature intercommunication, thus fostering a deeper comprehension of both spatial and spectral attributes. The LGIM equips the network with the capability to discern intricate land cover classifications by promoting an active interchange of local and global contextual insights, derived from CNN-extracted and transformer-extracted

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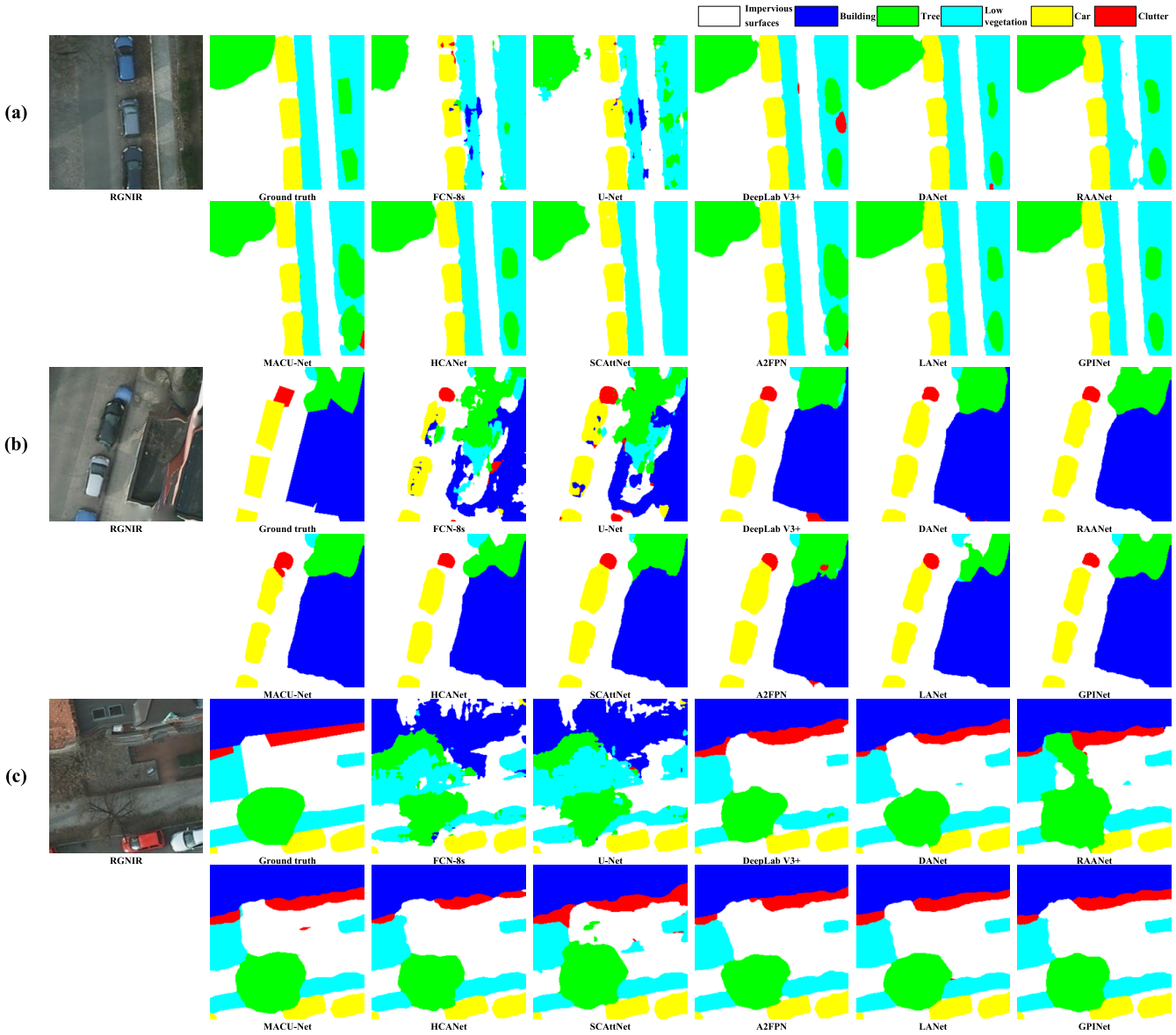


Fig. 6. Visualization comparisons on the ISPRS Potsdam test set.

TABLE V

RESULTS ON THE LOVEDA DATASET. CLASS-WISE $F1$-SCORE, AF, OA, AND MIOU ARE LISTED, WHERE THE BOLD TEXT INDICATES THE BEST RESULTS

methods	background	Building	Road	Water	Barren	Forest	Agriculture	AF	OA	mIoU
FCN-8s [12]	50.60	53.33	52.01	76.05	12.88	45.28	62.07	50.32	49.75	46.54
U-Net [15]	50.21	54.74	56.38	77.12	18.09	48.93	66.05	53.07	51.81	47.84
DeepLabV3+ [18]	52.29	54.99	57.16	77.96	16.11	48.18	67.79	53.50	52.30	47.62
DANet [27]	54.47	61.02	63.37	79.17	26.63	52.28	70.02	58.14	54.64	50.18
RAANet [63]	55.02	62.19	65.58	81.03	29.25	54.11	74.07	60.18	58.95	53.93
MACU-Net [64]	59.16	64.08	66.73	81.01	32.23	55.81	75.79	62.12	59.65	54.16
HCANet [47]	66.39	70.76	75.11	88.29	51.14	63.92	81.07	70.95	69.47	62.77
SCAttNet [45]	65.95	71.88	77.04	86.61	50.79	61.19	82.00	70.78	67.31	61.09
A2FPN [65]	65.17	73.32	75.19	88.01	48.82	59.96	79.71	70.03	66.89	61.14
LANet [43]	68.32	75.61	79.02	89.21	**53.23**	**66.02**	82.34	73.39	70.43	63.35
GPINet (ours)	**69.45**	**80.18**	**82.25**	**91.86**	53.16	65.87	**84.48**	**75.32**	**72.44**	**65.92**

features, respectively. In this section, we eliminated the LGIM from the interconnection between the transformer and CNN branches. Instead, we opted to merge the features at the end of the CNN and transformer processes via concatenation, succeeded by a 1×1 convolutional layer.

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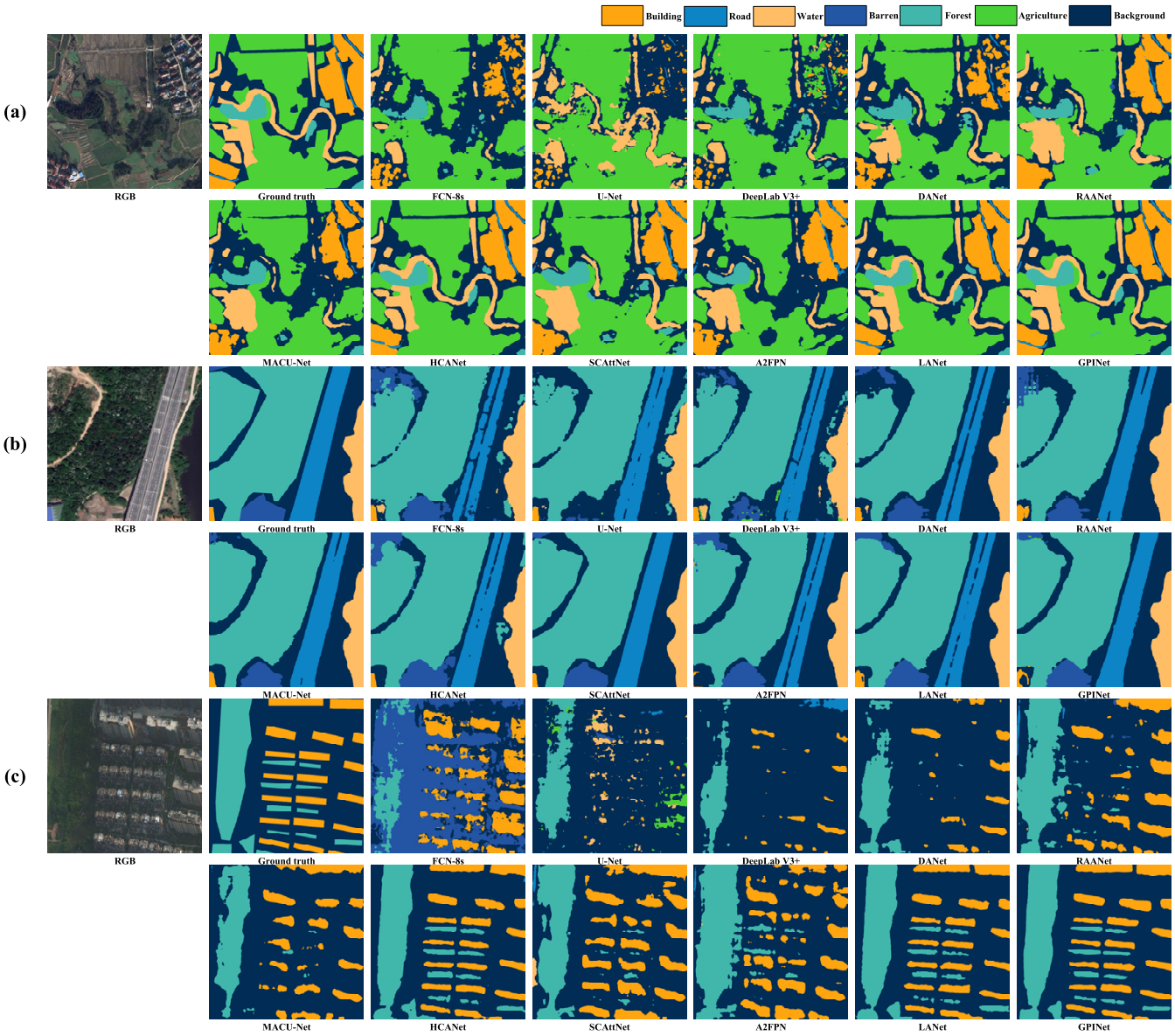


Fig. 7. Visualization comparisons on the LoveDA test set.

TABLE VI

EFFECTS OF LGIM ON THREE BENCHMARKS. RESULTS ARE IN THE FORM OF AF/OA/MIOU

Models	ISPRS Vaihingen	ISPRS Potsdam	LoveDA
GPINet	91.58/89.40/82.57	93.00/91.57/83.33	75.32/72.44/65.92
GPINet w/o LGIM	87.22/85.15/78.64	89.23/87.10/77.19	69.19/65.57/61.02

TABLE VII

EFFECTS OF DIFFERENT DECODERS ON THREE BENCHMARKS. RESULTS ARE IN THE FORM OF AF/OA/MIOU

Models	ISPRS Vaihingen	ISPRS Potsdam	LoveDA
GPINet	91.58/89.40/82.57	93.00/91.57/83.33	75.32/72.44/65.92
GPINet w/o PGD	86.19/82.14/75.91	88.45/84.09/76.52	69.93/67.26/61.21

Quantitative results from the ISPRS Vaihingen, ISPRS Potsdam, and LoveDA datasets—reported in Table VI—demonstrate marked improvements across all metrics with the integration of LGIM. Specifically, for the Vaihingen dataset, LGIM's inclusion correlates with a 4.36% increase in AF, a 4.25% rise in OA, and a 3.93% enhancement in mIoU. The Potsdam dataset reflects similar gains with improvements of 3.77% in AF, 4.47% in OA, and a notable 6.14% in mIoU.

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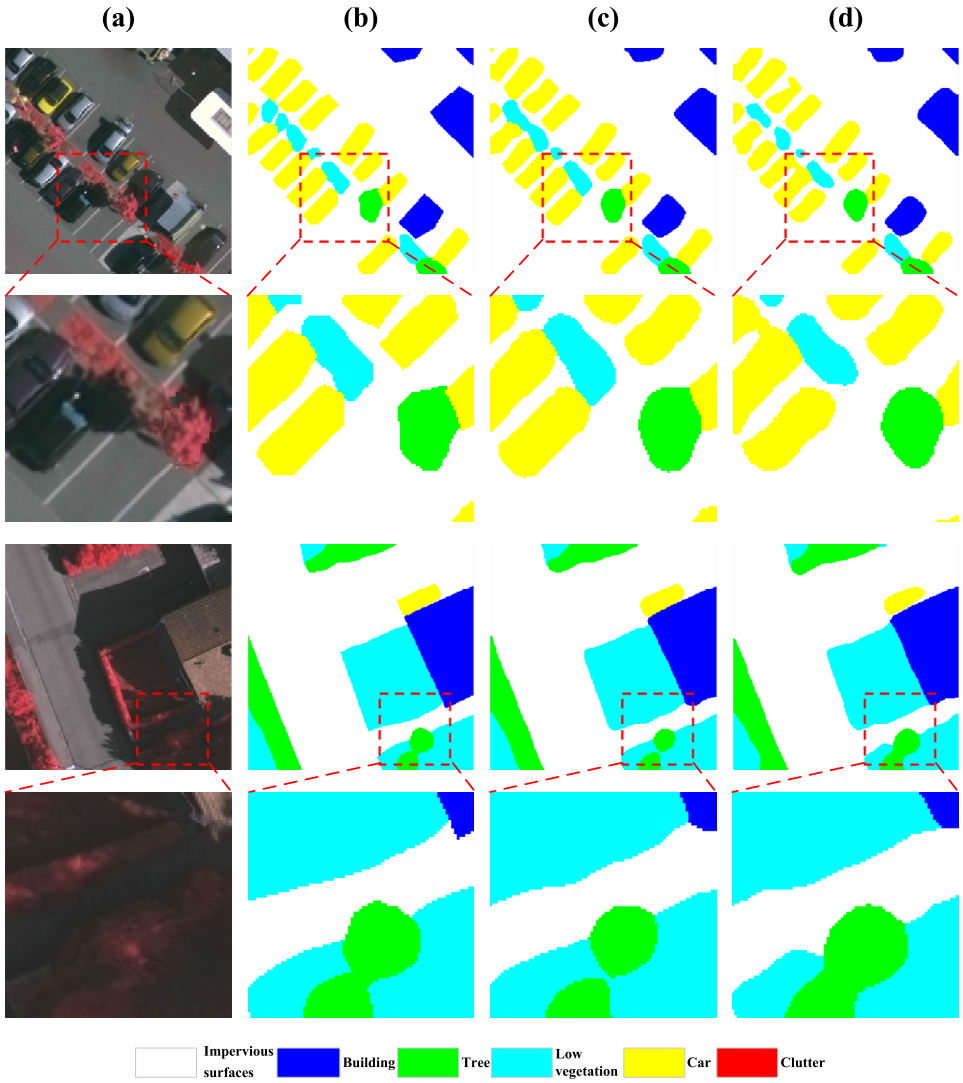


Fig. 8. Visualizations on the ISPRS Vaihingen test samples. (a) RGB image, (b) ground truth, (c) GPINet, and (d) GPINet w/o PGD.

The LoveDA dataset showcases the module's impact most significantly in mIoU, with a 4.9% improvement, accompanied by a 6.13% increase in AF and a 6.87% rise in OA.

These performance enhancements attributable to LGIM suggest its effectiveness in capturing local geometric features essential for the precise segmentation of complex images. The substantial increase in mIoU across the datasets indicates a more accurate pixel-level classification, leading to better class-specific segmentation. The OA metric further implies that LGIM enables the model to correctly classify a higher percentage of the total pixels, affirming the module's capability to reduce classification errors and refine spatial details. This is particularly crucial in environments with intricate geometric patterns, such as urban areas where the delineation of buildings, roads, and vegetation types can benefit greatly from the nuanced geometric understanding that LGIM provides. The consistent improvement in segmentation results, regardless of the landscape, emphasizes LGIM's robustness and its potential for generalization, reinforcing the argument for the inclusion of geometric context in deep learning models for RSIs.

F. Effects of PGD

We conducted an ablation study to verify the impact of geometric priors. Instead of utilizing PGD, we employed an upsampling layer (with a ratio of 32 in this study) to recover feature maps, which is referred to as GPINet w/o PGD. As reported in Table VII, the absence of geometric priors leads to a noticeable decrease in OA and mIoU by around 5%. We attribute this degradation to the crucial role of refining decoder features in enhancing the distinguishability of learned representations. The designed PGD incorporates geometric priors which offer reliable and informative contextual information. Consequently, incorporating rich context information makes the pixel-wise semantics more distinguishable. To sum up, the numerical metrics of GPINet show significant improvements compared to the alternative models.

Figs. 8–10 show the segmentation results for the ISPRS Vaihingen, Potsdam, and LoveDA test samples. The implementation of the PGD within the GPINet framework has shown a profound impact on the segmentation accuracy

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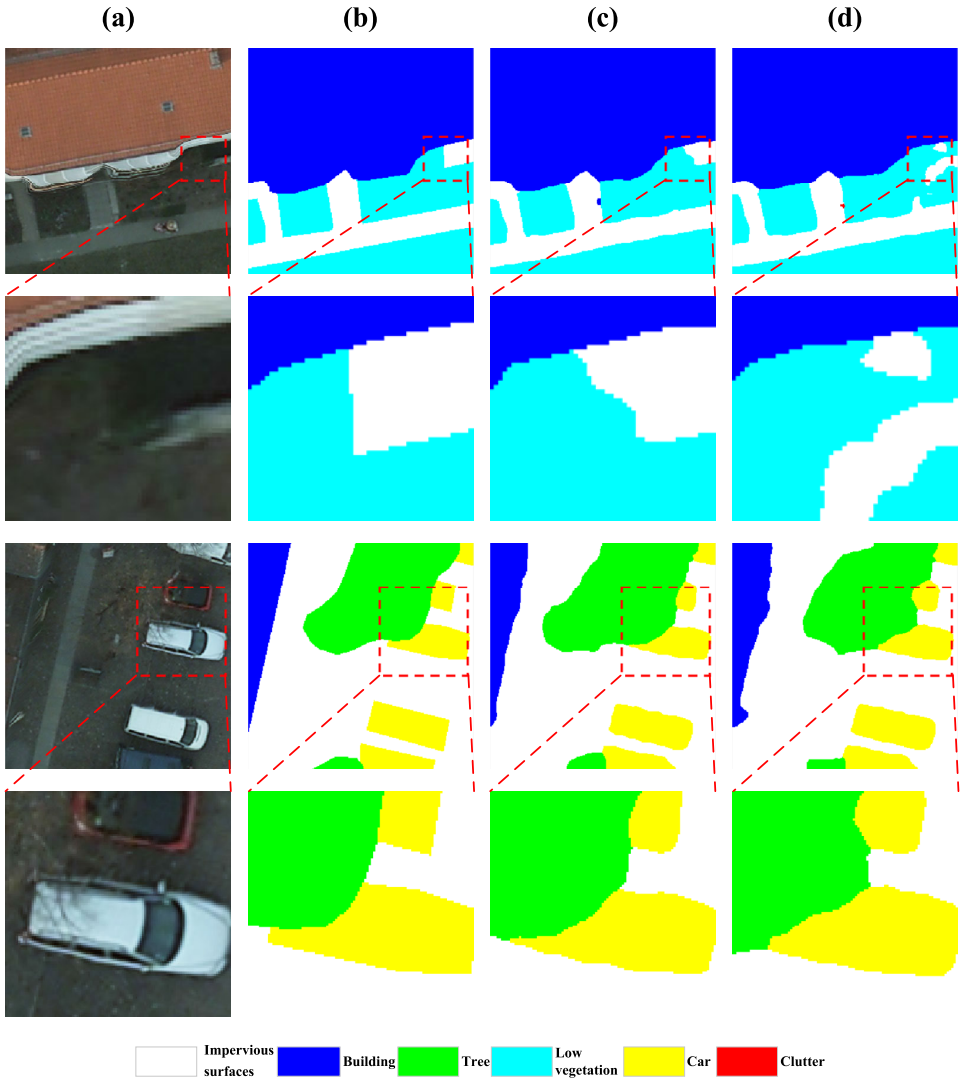


Fig. 9. Visualizations on the ISPRS Potsdam test samples. (a) RGB image, (b) ground truth, (c) GPINet, and (d) GPINet w/o PGD.

of various datasets, particularly in delineating complex urban structures and vegetation. In the Vaihingen dataset, the application of PGD has led to an appreciable refinement in the depiction of geometric details, which is evident in the enhanced delineation of building edges, tree canopies, and the separation of overlapping features. Similarly, in the Potsdam dataset, the PGD has markedly improved the segmentation of intricate urban elements such as vehicles and complex building layouts, which are typically challenging for segmentation models that do not utilize geometric priors.

The LoveDA dataset, with its unique challenges posed by diverse urban and rural landscapes, further underscores the efficacy of PGD. The geometric priors employed by PGD facilitate the distinction between natural and artificial structures, allowing for a more nuanced interpretation of the landscape. This is particularly evident in the accurate classification of rural features such as small farm plots and irrigation canals, which often share spectral similarities with urban elements but differ significantly in geometric structure.

Instances, where PGD's influence is most conspicuous, include the segmentation of roundabouts, the consistency in identifying narrow roads amidst dense vegetation, and the delineation of small water bodies surrounded by urban development. These instances demonstrate the strength of PGD in capturing geometric consistency and enhancing the segmentation model's robustness against varied and complex backgrounds.

The results across these datasets reveal that the incorporation of geometric priors through PGD not only improves the model's ability to understand the spatial context but also its capability to generalize across different environments. By doing so, the GPINet with PGD provides a versatile solution to one of the persistent challenges in remote sensing: the accurate, fine-grained segmentation of heterogeneous landscapes. This capability is critical in applications such as urban planning, environmental monitoring, and land use

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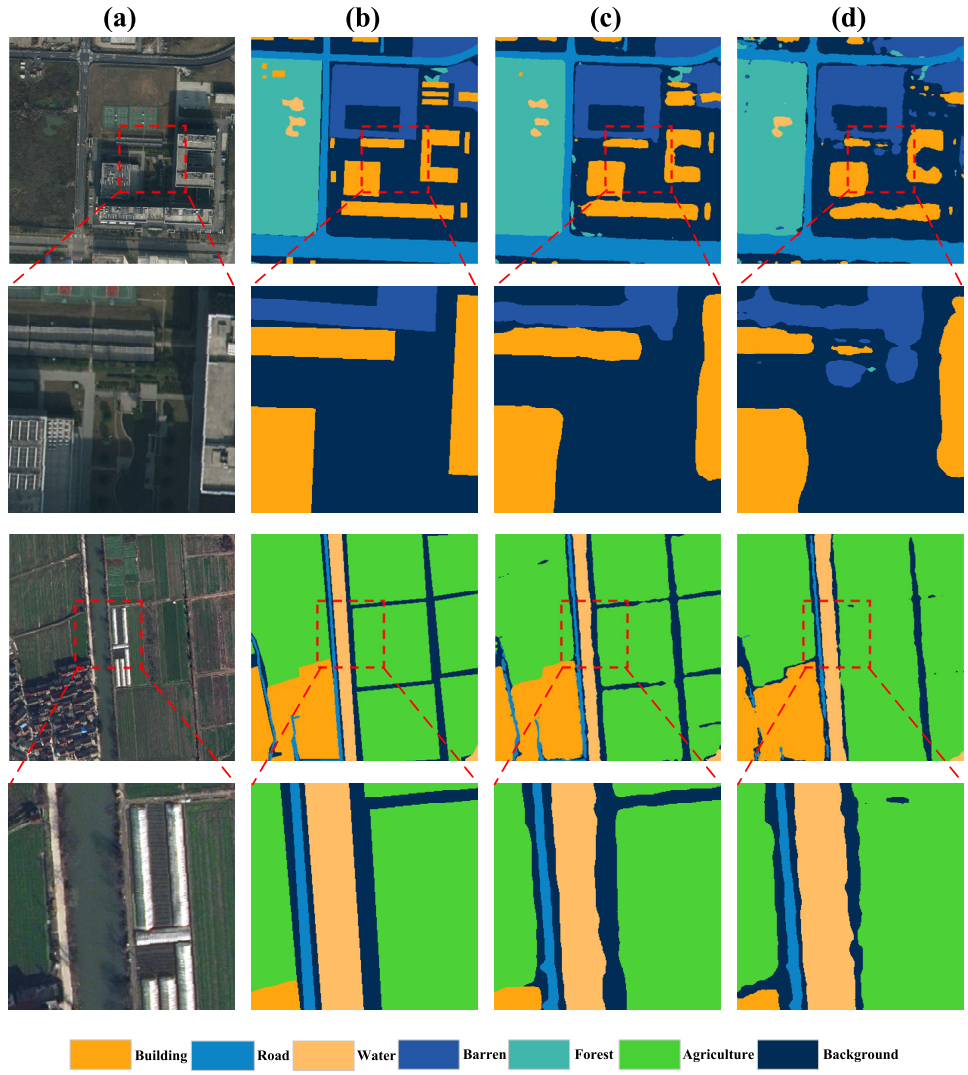


Fig. 10. Visualizations on the LoveDA test samples. (a) RGB image, (b) ground truth, (c) GPINet, and (d) GPINet w/o PGD.

classification, where the precise delineation of ground objects is paramount.

G. Efficiency Analysis

In the context of efficiency analysis, we compared the model size with an input size of $256 \times 256 \times 3$. To ensure a fair comparison, we set the batch size to 1 for testing. As shown in Table VIII, GPINet has a relatively small number of parameters. Notably, GPINet achieves better results than FCN-8s, U-Net, DeepLab V3+, DANet, RAANet, HCANet, and A2FPN despite having fewer parameters. Moreover, when compared to the previous best model, LANet, GPINet only has a slightly larger number of parameters within 1%, yet it outperforms LANet in terms of OA by more than 1% on all three benchmarks. In summary, the proposed GPINet reaches a state-of-the-art accuracy with a relatively small number of parameters.

Table VIII presents the calculations for FLOPs and inference times (with an input size of $256 \times 256 \times 3$). Compared with baseline models like FCN-8s, U-Net, DeepLab V3+, and DANet, GPINet demonstrates lower FLOPs and inference times, all while maintaining high accuracy. When contrasted with LANet, our approach exhibits slightly reduced efficiency. However, this minor efficiency disparity is inconsequential when weighed against its accuracy performance. These observations hold true across various metrics, including inference times. GPINet strikes a commendable balance between accuracy and efficiency in the realm of semantic segmentation for RSIs.

H. Convergence Comparison

We assessed the training loss convergence of several models, including RAANet, MACU-Net, HCANet, SCAttNet, A2FPN, and LANet, across three datasets. In Fig. 11, the convergence curves for the ISPRS Vaihingen dataset are depicted. GPINet consistently maintains the lowest cross-entropy loss during training. Remarkably, GPINet achieves stable convergence within just 100 epochs, whereas the other models require over 200 epochs to do the same. Similarly, Fig. 12 shows the rapid convergence performance of GPINet when trained

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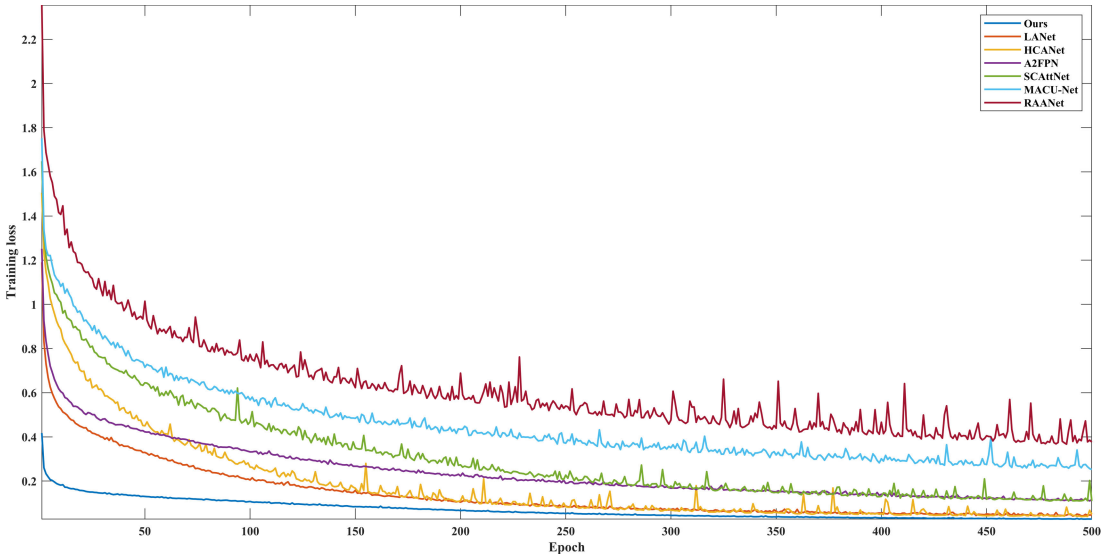


Fig. 11. Training loss of ISPRS Vaihingen dataset.

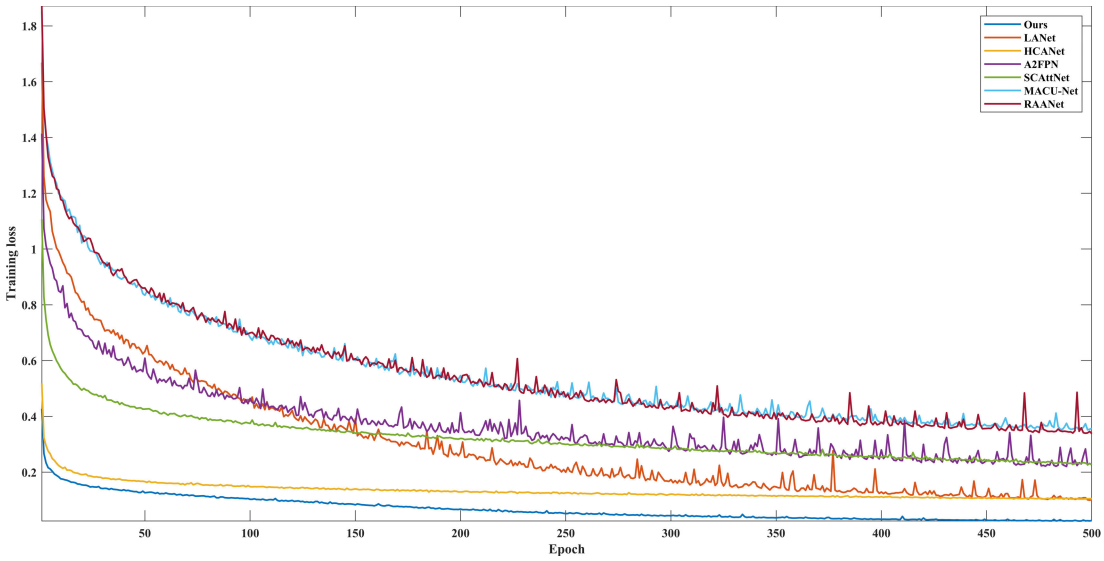


Fig. 12. Training loss of ISPRS Potsdam dataset.

TABLE VIII
EFFICIENCY COMPARISONS

Methods	Parameters (M)	FLOPs(G)	Inference time (ms)
FCN-8s	29.78	27.50	19.81
U-Net	31.06	28.68	20.66
DeepLabV3	39.41	36.39	26.21
DANet	48.22	44.53	32.07
RAANet	79.50	73.14	52.87
MACU-Net	5.15	4.75	3.73
HCANet	42.25	39.02	28.10
SCAttNet	26.62	24.58	17.70
A2FPN	39.98	36.92	26.59
LANet	23.78	21.96	15.82
GPINet (ours)	27.70	23.67	18.42

on the ISPRS Potsdam dataset. For satellite images, as shown in Fig. 13, GPINet converges in 200 epochs, while the other models consistently demand 350 epochs for convergence. It is noteworthy that across all three datasets, GPINet consistently converges with the lowest loss, demonstrating its robust capacity to learn meaningful and contributory features.

I. Generalization Ability Analysis

In this section, we discuss the generalization ability of GPINet. Prior to conducting experiments, the design of GPINet is focused on incorporating comprehensive contexts in both spatial and spectral domains. Regarding the geometric properties of RSIs, PGD is devised and embedded to replace the standard decoder, which lacks the constraints of geometric priors. Then, our selection of benchmark datasets, including ISPRS Vaihingen, ISPRS Potsdam, and LoveDA, aims to encompass a wide range of spatial resolutions, geographical regions, and sensor characteristics commonly encountered in real-world RSIs. The consistent and competitive performance of GPINet across these datasets reflects its robustness in handling the intricate details and spectral variations inherent

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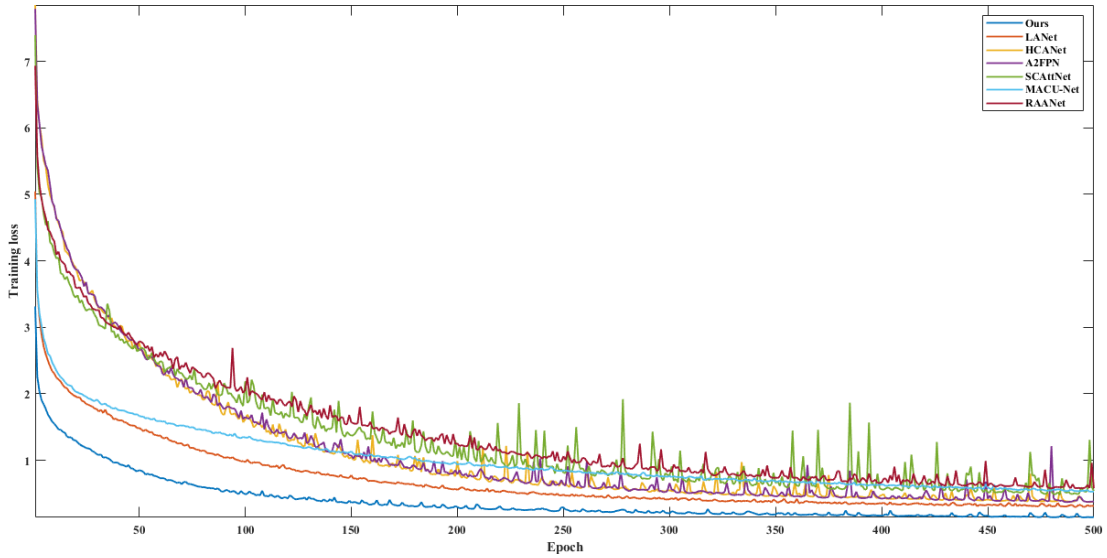


Fig. 13. Training loss of LovaDA dataset.

in RSIs. Collectively, our rigorous evaluation demonstrates the model's remarkable generalization ability, making it a valuable tool for diverse practical remote sensing applications.

V. Conclusion

In this study, we propose a hybrid GPINet for the semantic segmentation of RSIs. In the encoder stage, a dual branch structure with a specific interaction module is designed for coupling and exchanging local and global contexts. This design enables the network to refine learned representations interactively and efficiently. Due to the ubiquitous scale, spectrum, and appearance variations, we propose to store and update the geometric features iteratively. These geometric priors are consistent with class-level semantic masks and can provide credible guidance for feature decoding and inference. Experiments on three benchmark datasets demonstrate that GPINet outperforms the state-of-the-art methods with comparable parameters. In summary, our results provide compelling evidence that comprehensive contexts with geometric prior guidance are valuable for semantic segmentation of RSIs. On the one hand, we plan to explore the impact of varying noise levels on our method's performance in the semantic segmentation of RSIs. This investigation will contribute to a more comprehensive understanding of GPINet's robustness in real-world scenarios. On the other hand, future efforts to extend this work may include the exploration of a more effective combination of CNNs and transformers while preserving informative features, and the development of an advanced CNN-transformer network for few-shot semantic segmentation of RSIs to tackle the challenge of insufficient samples.

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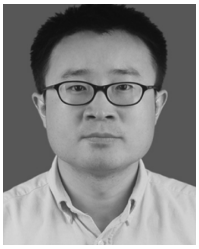


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